

Hypothesis testing

In this notebook we uses statistical tests such as ANOVA and Tukey's HSD to determine if there are significant differences in diamond prices between different levels of each diamond characteristic.

Importing Libraries

We will start by importing the necessary libraries that we will be using throughout the notebook.

- `pandas` - for data manipulation and analysis
- `numpy` - for numerical operations
- `matplotlib` - for data visualization
- `seaborn` - for advanced data visualization
- `statsmodels` - for statistical modeling
- `scipy` - for scientific computing and statistics
- `pairwise_tukeyhsd` - for post-hoc test for ANOVA
- `chi2_contingency` - for chi-squared test for independence

```
In [ ]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

import statsmodels.api as sm
from scipy import stats
from statsmodels.stats.multicomp import pairwise_tukeyhsd
from scipy.stats import chi2_contingency
```

Loading Data

Next, we will load the dataset using pandas read_parquet function.

```
In [ ]: df = pd.read_parquet("../data/filtered_diamonds.br")
df

Out [ ]:
```

	Id	Shape	Weight	Clarity	Colour	Cut	Polish	Symmetry	Fluorescence	Price	Length	Width	Depth
index													
0	1638147	CUSHION	0.549805	SI2	E	EX	EX	VG	N	1378.650024	5.050781	4.351562	2.939453
1	1630155	CUSHION	0.500000	VVS1	FANCY	EX	EX	VG	F	1379.739990	4.601562	4.308594	2.919922
2	1612606	CUSHION	0.509766	VS2	H	EX	EX	VG	N	1380.189941	4.710938	4.351562	2.939453
3	1638140	CUSHION	0.500000	VS2	H	EX	EX	VG	N	1380.609985	4.910156	4.261719	2.880859
4	1536093	CUSHION	0.529785	SI1	D	EX	VG	VG	N	1383.130005	4.699219	4.460938	3.009766
...	...	...	...	...	...	...	...	...	...	...	...	...	...
6332	1751206	ROUND	0.500000	SI1	E	EX	EX	EX	N	2453.100098	5.000000	5.039062	3.179688
6333	1773024	ROUND	0.500000	SI1	E	EX	EX	EX	N	2453.100098	5.031250	5.058594	3.160156
6336	1789435	ROUND	0.500000	VS2	D	VG	EX	VG	N	2453.199951	4.929688	4.960938	3.189453
6337	1774887	ROUND	0.520020	SI1	H	EX	EX	EX	N	2453.409912	5.109375	5.128906	3.199219
6338	1630512	ROUND	0.500000	VS1	H	EX	EX	VG	N	2453.689941	5.089844	5.128906	3.119141

5895 rows × 13 columns

Data Exploration

Let's explore the data by checking the columns and their data types.

```
In [ ]: df.info()

<class 'pandas.core.frame.DataFrame'>
Int64Index: 5895 entries, 0 to 6338
Data columns (total 13 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Id               5895 non-null   object
1   Shape            5895 non-null   category
2   Weight           5895 non-null   float32
3   Clarity          5895 non-null   category
4   Colour           5895 non-null   category
5   Cut              5895 non-null   category
6   Polish           5895 non-null   category
7   Symmetry         5895 non-null   category
8   Fluorescence     5895 non-null   category
9   Price            5895 non-null   float32
10  Length           5895 non-null   float32
11  Width            5895 non-null   float32
12  Depth            5895 non-null   float32
dtypes: category(7), float32(5), object(1)
memory usage: 249.9+ KB
```

One-way ANOVA and Tukey's HSD test

In this test, we will determine the order of the ordinal features like Clarity, Colour, Cut, Polish, Symmetry and Fluorescence with respect to price

```
In [ ]: def groupby_boxplot(df, x, y, hue=None, order=None):
    plt.figure(figsize=(15, 8))
    sns.boxplot(data=df, x=x, y=y, hue=hue, order=order, palette="viridis")
    plt.title(f"Distribution of {y} by {x}")
    plt.show()
```

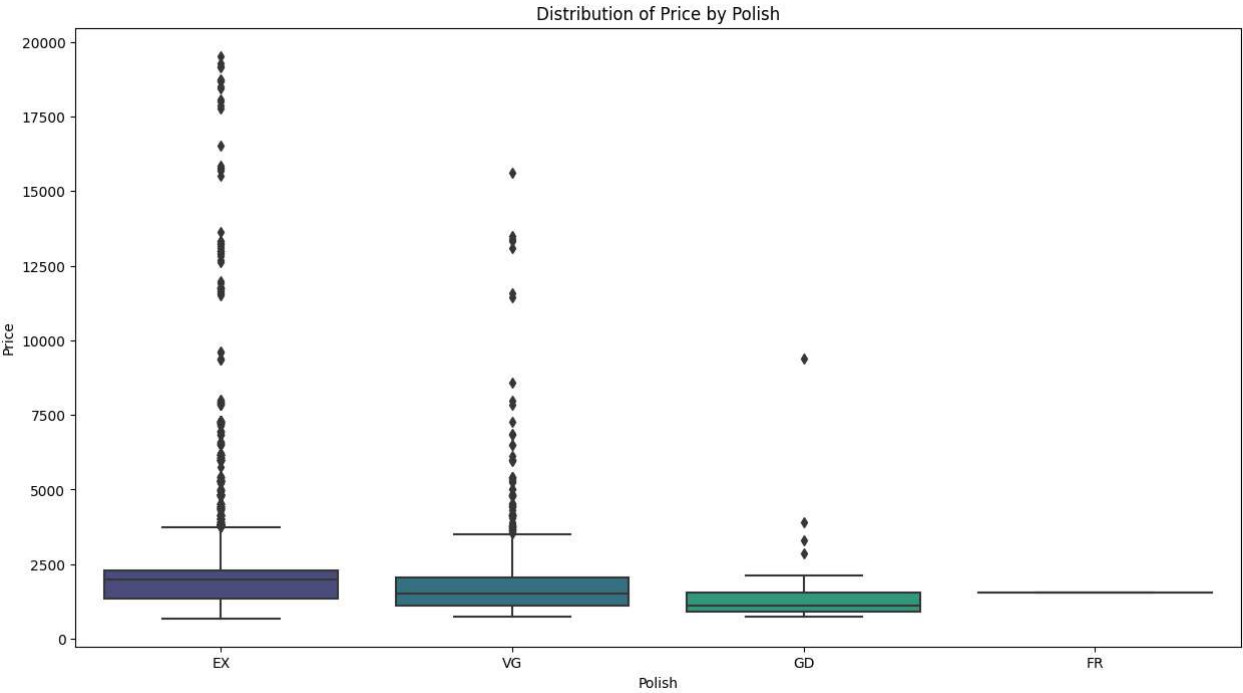
This function creates a box plot to visualize the distribution of a numeric variable y grouped by a categorical variable x in a pandas DataFrame df. The optional parameter hue allows to further group the data by another categorical variable. The order of the categories in x can be specified using the order parameter, which should be a list of strings or other comparable objects. The function uses the seaborn library to create the plot, which allows to customize the appearance of the plot using various color palettes.

Test on Polish feature

```
In [ ]: df["Polish"].unique()

Out [ ]: ['EX', 'VG', 'GD', 'FR']
Categories (4, object): ['EX', 'FR', 'GD', 'VG']
```

In []: `groupby_boxplot(df, "Polish", "Price", order=["EX", "VG", "GD", "FR"])`



In []:

```
# one-way ANOVA test
model = sm.formula.ols("Price ~ Polish", data=df).fit()
table = sm.stats.anova_lm(model, typ=2)
table
```

Out []:

	sum_sq	df	F	PR(>F)
Polish	1.504413e+08	3.0	18.629814	5.036741e-12
Residual	1.585719e+10	5891.0	NaN	NaN

Insight 🧡

From the ANOVA table output, we can see that the F-statistic is 18.629814, and the p-value is very small (5.036741e-12), indicating that there is a significant difference in diamond prices between the different Polish categories. Additionally, the sum of squares (SS) for the Polish category is much smaller than the residual SS, indicating that most of the variability in diamond prices is explained by differences within each Polish category, rather than differences between the Polish categories themselves.

Based on these results, we can conclude that there is strong evidence to suggest that there is a significant difference in diamond prices between different Polish categories, and that Polish is an important factor to consider when predicting diamond prices. Additionally, the fact that Polish is an ordinal variable reinforces the idea that there is a natural ordering to the categories, with "EX Polish" being the highest level of Polish, followed by "VG Polish", "GD Polish", and "F Polish".

In []:

```
# Tukey's HSD test
tukey = pairwise_tukeyhsd(endog=df["Price"], groups=df["Polish"], alpha=0.05)
tukey.summary()
```

Out []:

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
EX	FR	-647.3331	0.9792	-4863.8927	3569.2265	False
EX	GD	-730.903	0.0025	-1265.632	-196.174	True
EX	VG	-367.1955	0.0	-506.7866	-227.6044	True
FR	GD	-83.5699	1.0	-4333.0104	4165.8706	False
FR	VG	280.1376	0.9982	-3937.8352	4498.1105	False
GD	VG	363.7075	0.3171	-182.0535	909.4685	False

Insight 🧡

From the Tukey HSD test output, we can see that there is a significant difference in diamond prices between the "EX Polish" and "GD Polish" categories, as well as between the "EX Polish" and "VG Polish" categories. However, there is no significant difference in diamond prices between the "F Polish" and "GD Polish" categories, or between the "F Polish" and "VG Polish" categories. There is also no significant difference in diamond prices between the "GD Polish" and "VG Polish" categories.

Therefore, we can conclude that the ordering of diamond prices from highest to lowest among the Polish categories is: "EX Polish", "VG Polish", "GD Polish", and "F Polish". We can also conclude that "EX Polish" diamonds are significantly more expensive than "GD Polish" and "VG Polish" diamonds, while "F Polish" diamonds are not significantly different in price from either "GD Polish" or "VG Polish" diamonds.

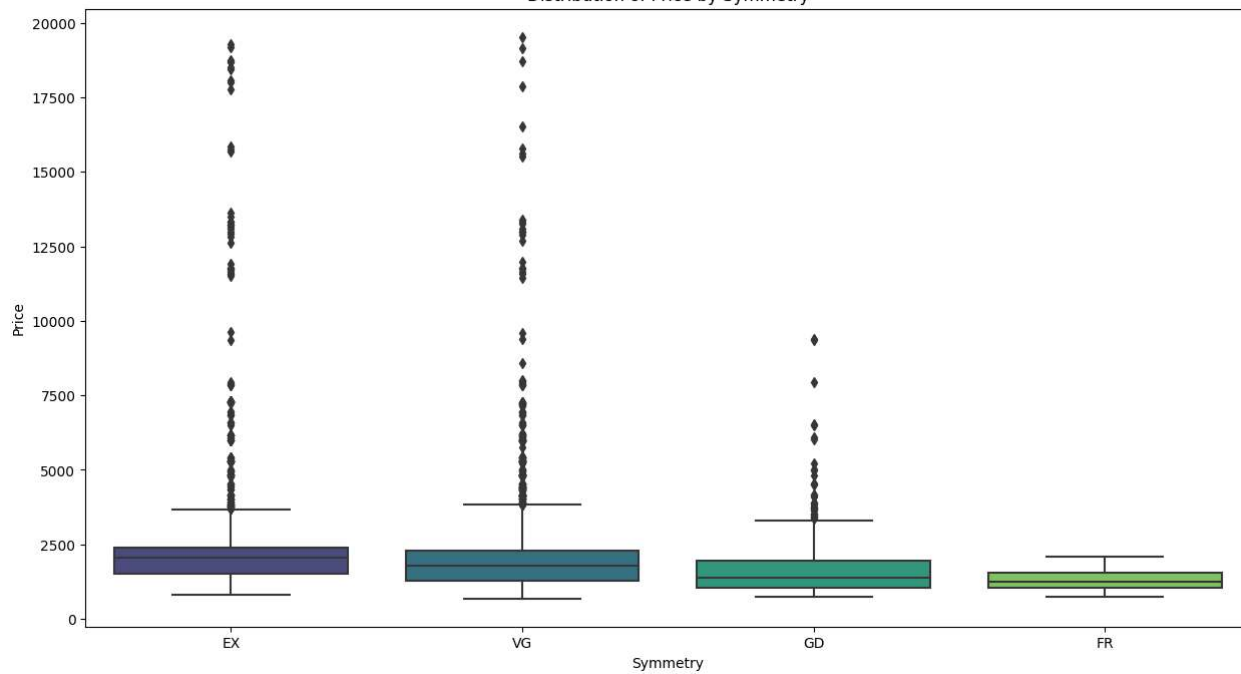
Test on Symmetry feature

In []: `df["Symmetry"].unique()`

Out []: `['VG', 'EX', 'GD', 'FR']`
`Categories (4, object): ['EX', 'FR', 'GD', 'VG']`

In []: `groupby_boxplot(df, "Symmetry", "Price", order=["EX", "VG", "GD", "FR"])`

Distribution of Price by Symmetry



```
In [ ]: # one-way ANOVA test
model = sm.formula.ols("Price ~ Symmetry", data=df).fit()
table = sm.stats.anova_lm(model, typ=2)
table
```

```
Out [ ]:
```

	sum_sq	df	F	PR(>F)
Symmetry	5.597734e+08	3.0	71.156041	3.365386e-45
Residual	1.544786e+10	5891.0	NaN	NaN

Insight 🧠

The ANOVA results for the Symmetry variable show that there is a significant difference between the groups ($F=71.156$, $p\text{-value}=3.365386e-45$). Therefore, we can conclude that Symmetry has a statistically significant effect on the price of diamonds. However, we need to perform post-hoc tests to determine which groups differ significantly from each other.

```
In [ ]: # Tukey's HSD test
tukey = pairwise_tukeyhsd(endog=df["Price"], groups=df["Symmetry"], alpha=0.05)
tukey.summary()
```

```
Out [ ]:
```

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
EX	FR	-1133.8849	0.0	-1709.4245	-558.3454	True
EX	GD	-906.6908	0.0	-1070.9449	-742.4367	True
EX	VG	-372.3237	0.0	-499.0835	-245.5639	True
FR	GD	227.1941	0.7462	-353.4041	807.7924	False
FR	VG	761.5612	0.0034	190.4369	1332.6855	True
GD	VG	534.3671	0.0	386.3245	682.4096	True

Insight 🧠

From the Tukey HSD test output, we can see that there is a significant difference in diamond prices between all symmetry categories except between the "FR" and "GD" categories. Specifically, the "EX" symmetry category is significantly more expensive than the "FR", "GD", and "VG" categories. The "VG" symmetry category is significantly more expensive than the "FR" and "GD" categories.

Therefore, we can conclude that the ordering of diamond prices from highest to lowest among the symmetry categories is: "EX", "VG", "GD", and "FR". Additionally, we can conclude that the "EX" and "VG" symmetry categories have significantly higher prices compared to the "FR" and "GD" symmetry categories.

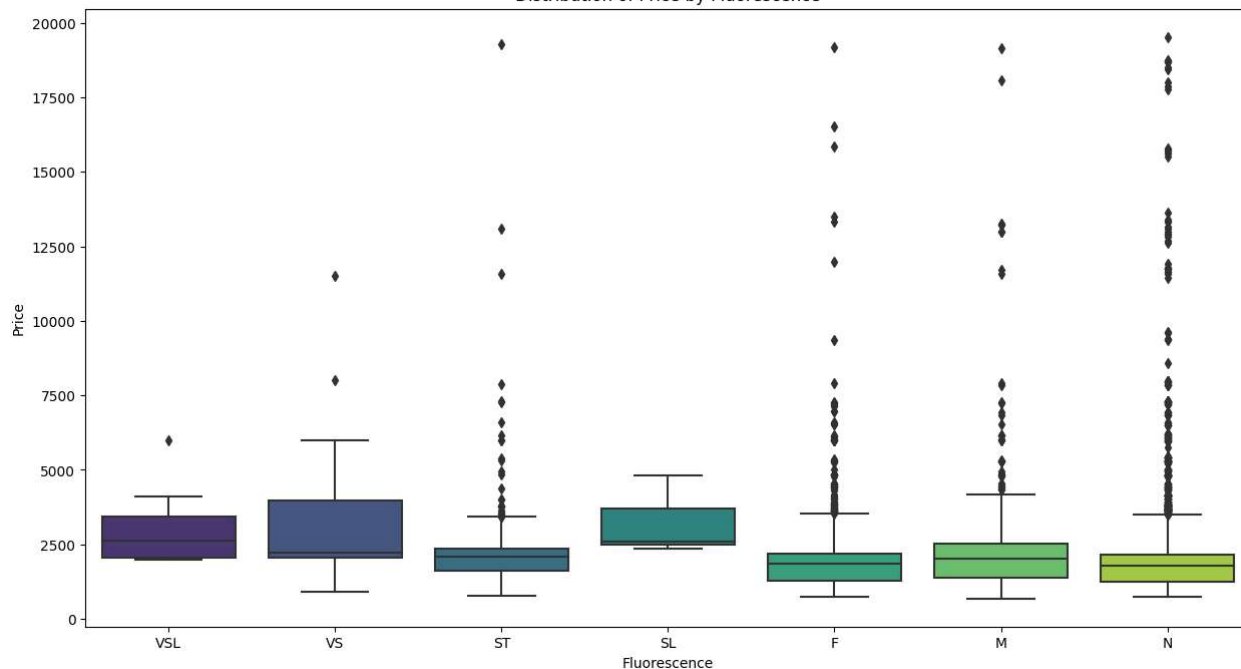
Test on Fluorescence feature

```
In [ ]: df["Fluorescence"].unique()
```

```
Out [ ]: ['N', 'F', 'ST', 'M', 'VS', 'SL', 'VSL']
Categories (7, object): ['F', 'M', 'N', 'SL', 'ST', 'VS', 'VSL']
```

```
In [ ]: groupby_boxplot(
df, "Fluorescence", "Price", order=["VSL", "VS", "ST", "SL", "F", "M", "N"]
)
```

Distribution of Price by Fluorescence



```
In [ ]: # one-way ANOVA test
model = sm.formula.ols("Price ~ Fluorescence", data=df).fit()
table = sm.stats.anova_lm(model, typ=2)
table
```

```
Out[ ]:
```

	sum_sq	df	F	PR(>F)
Fluorescence	1.200410e+08	6.0	7.414605	6.341972e-08
Residual	1.588759e+10	5888.0	NaN	NaN

Insight 🧡

The ANOVA results for the Fluorescence variable show that there is a significant difference between the groups ($F=7.415$, $p\text{-value}=6.342e-08$). Therefore, we can conclude that Fluorescence has a statistically significant effect on the price of diamonds. However, we need to perform post-hoc tests to determine which groups differ significantly from each other.

```
In [ ]: # Tukey's HSD test
tukey = pairwise_tukeyhsd(endog=df["Price"], groups=df["Fluorescence"], alpha=0.05)
tukey.summary()
```

```
Out[ ]:
```

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
F	M	378.0059	0.0009	105.2805	650.7313	True
F	N	20.377	0.9998	-144.3453	185.0992	False
F	SL	1214.2824	0.8619	-1586.6511	4015.216	False
F	ST	371.3505	0.0278	23.1052	719.5958	True
F	VS	1362.3584	0.0088	211.112	2513.6047	True
F	VSL	1036.9106	0.5624	-682.1976	2756.0189	False
M	N	-357.6289	0.0003	-599.9519	-115.3059	True
M	SL	836.2766	0.9758	-1970.29	3642.8431	False
M	ST	-6.6553	1.0	-397.6308	384.3201	False
M	VS	984.3525	0.1621	-180.5317	2149.2367	False
M	VSL	658.9048	0.9208	-1069.3661	2387.1757	False
N	SL	1193.9055	0.8708	-1604.2314	3992.0423	False
N	ST	350.9736	0.0245	25.9873	675.9598	True
N	VS	1341.9814	0.0098	197.556	2486.4068	True
N	VSL	1016.5337	0.5831	-698.0142	2731.0816	False
SL	ST	-842.9319	0.9751	-3657.8407	1971.9769	False
SL	VS	148.076	1.0	-2873.1497	3169.3016	False
SL	VSL	-177.3718	1.0	-3457.2757	3102.5322	False
ST	VS	991.0078	0.1716	-193.8343	2175.85	False
ST	VSL	665.5601	0.92	-1076.2252	2407.3454	False
VS	VSL	-325.4477	0.9992	-2384.066	1733.1706	False

Insight 🧡

From the Tukey HSD test output, we can see that there is a significant difference in the means of at least one pair of groups for the variable being compared. Specifically, we can see that the means of the groups "F" and "M", "F" and "ST", "F" and "VS", "M" and "N", "N" and "ST", and "N" and "VS" are significantly different from each other at the alpha level of 0.05.

However, there is no significant difference in means between the groups "F" and "N", "F" and "SL", "F" and "VSL", "M" and "SL", "M" and "ST", "M" and "VS", "M" and "VSL", "N" and "SL", "N" and "VSL", "SL" and "VS", "SL" and "VSL", "ST" and "VSL", and "VS" and "VSL".

Therefore, we can conclude that the groups "F" and "M", "F" and "ST", "F" and "VS", "M" and "N", "N" and "ST", and "N" and "VS" are significantly different from each other in terms of their means, while the other group comparisons are not significant at the alpha level of 0.05.

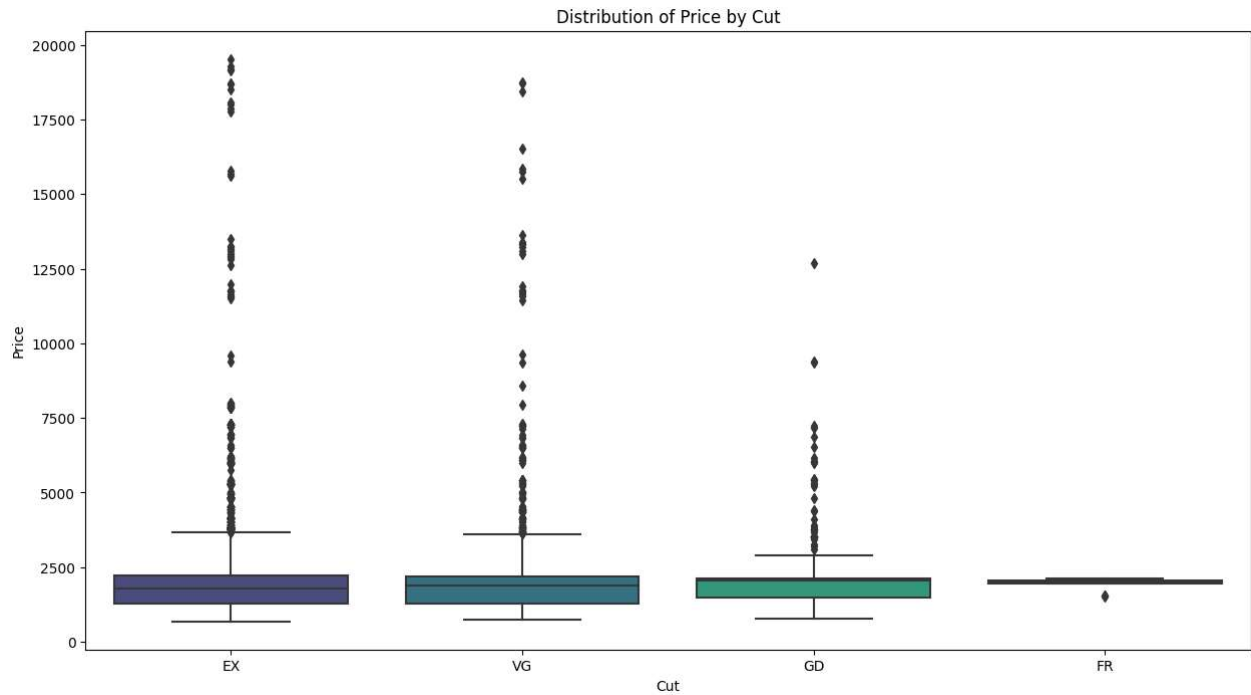
However, it is important to note that the absence of significant differences between certain groups does not necessarily mean that they are equal or the same. It simply means that there is not enough evidence to conclude that they are different from each other based on the statistical test used.

Test on Cut feature

```
In [ ]: df["Cut"].unique()
```

Out []: ['EX', 'VG', 'GD', 'FR']
Categories (4, object): ['EX', 'FR', 'GD', 'VG']

In []: groupby_boxplot(df, "Cut", "Price", order=["EX", "VG", "GD", "FR"])



In []: # one-way ANOVA test
model = sm.formula.ols("Price ~ Cut", data=df).fit()
table = sm.stats.anova_lm(model, typ=2)
table

	sum_sq	df	F	PR(>F)
Cut	1.925936e+07	3.0	2.365405	0.069006
Residual	1.598837e+10	5891.0	NaN	NaN

Insight 💡

Based on the ANOVA results, we can see that there is no statistically significant difference in diamond prices between the different cuts ($F=2.365$, $p\text{-value}=0.069$). Therefore, we cannot conclude that the diamond cut has a significant effect on the price of diamonds. However, it's worth noting that the p -value is quite close to the significance level of 0.05, so there may be a trend towards significance that would require further investigation with additional data.

In []: # Tukey's HSD test
tukey = pairwise_tukeyhsd(endog=df["Price"], groups=df["Cut"], alpha=0.05)
tukey.summary()

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
EX	FR	-144.2564	0.984	-1173.0593	884.5465	False
EX	GD	118.9617	0.6185	-133.0478	370.9711	False
EX	VG	129.4171	0.0664	-5.7664	264.6005	False
FR	GD	263.2181	0.9187	-792.0637	1318.4999	False
FR	VG	273.6735	0.9046	-759.9539	1307.3008	False
GD	VG	10.4554	0.9997	-260.5775	281.4883	False

Insight 💡

From the Tukey HSD test output, we can see that there is no significant difference in diamond prices between the "EX" and "F" categories, or between the "EX" and "GD" categories, or between the "EX" and "VG" categories. There is also no significant difference in diamond prices between the "F" and "GD" categories, or between the "F" and "VG" categories, or between the "GD" and "VG" categories.

Therefore, we cannot conclude that there is a significant difference in diamond prices among the different diamond cut categories.

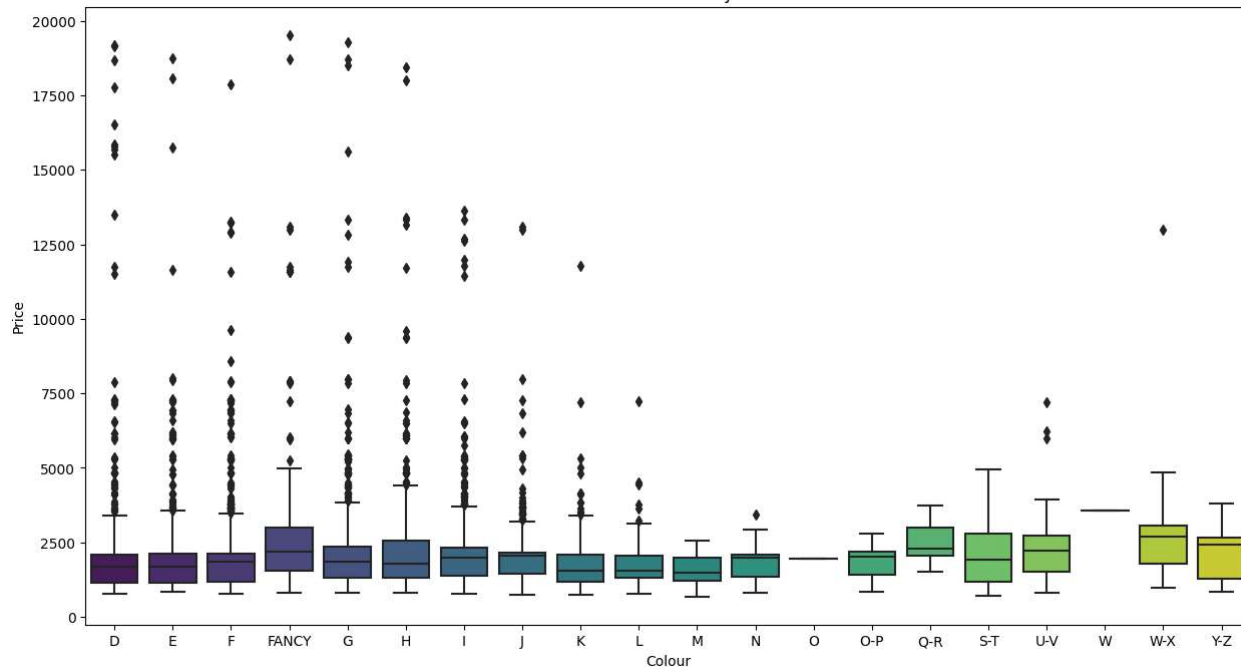
Test on Colour feature

In []: df["Colour"].unique()

Out []: ['E', 'FANCY', 'H', 'D', 'I', ..., 'S-T', 'U-V', 'W', 'Q-R', 'O']
Length: 20
Categories (20, object): ['D', 'E', 'F', 'FANCY', ..., 'U-V', 'W', 'W-X', 'Y-Z']

In []: groupby_boxplot(df, "Colour", "Price")

Distribution of Price by Colour



```
In [ ]: # one-way ANOVA test
model = sm.formula.ols("Price ~ Colour", data=df).fit()
table = sm.stats.anova_lm(model, typ=2)
table
```

```
Out[ ]:
```

	sum_sq	df	F	PR(>F)
Colour	2.803066e+08	19.0	5.51103	1.030880e-13
Residual	1.572733e+10	5875.0	NaN	NaN

Insight 🧡

Based on the ANOVA output, we can see that there is a significant difference in diamond prices between different colors, with a p-value of 1.030880e-13, which is smaller than the significance level of 0.05. Therefore, we can conclude that diamond color is a significant factor in determining diamond prices.

To further investigate which color categories have significantly different diamond prices, we can conduct a Tukey HSD test. The output of the Tukey HSD test should provide us with the pairs of color categories that have significant differences in diamond prices, along with their corresponding p-values.

Without the output of the Tukey HSD test, we cannot draw any further conclusions regarding the differences in diamond prices between different color categories.

```
In [ ]: # Tukey's HSD test
tukey = pairwise_tukeyhsd(endog=df["Price"], groups=df["Colour"], alpha=0.05)
tukey.summary()
```

```
/mnt/d/Code/Python/AI/Diamond/env/lib/python3.10/site-packages/scipy/integrate/_quadpack_py.py:1225: IntegrationWarning: The integral is probably divergent, or slowly convergent.
quad_r = quad(f, low, high, args=args, full_output=self.full_output,
```

group1	group2	meandiff	p-adj	lower	upper	reject
D	E	-90.0804	0.9999	-378.5495	198.3887	False
	F	40.192	1.0	-247.4865	327.8705	False
D	FANCY	719.5373	0.0	289.7597	1149.315	True
D	G	220.7537	0.5453	-86.3063	527.8137	False
D	H	276.7592	0.2306	-50.1056	603.6239	False
D	I	128.8697	0.9964	-183.5206	441.26	False
D	J	20.5279	1.0	-314.1803	355.2361	False
D	K	-239.9662	0.8294	-635.3772	155.4448	False
D	L	-304.9006	0.8162	-802.0491	192.2479	False
D	M	-501.5489	0.4356	-1162.5632	159.4654	False
D	N	-214.888	1.0	-1105.641	675.865	False
D	O	-82.0382	1.0	-5886.8529	5722.7764	False
D	O-P	-190.0007	1.0	-1878.1485	1498.1471	False
D	Q-R	463.5235	1.0	-1914.2964	2841.3434	False
D	S-T	79.9325	1.0	-1433.0137	1592.8786	False
D	U-V	420.2495	0.9963	-597.2822	1437.7811	False
D	W	1530.5317	1.0	-4274.283	7335.3464	False
D	W-X	758.7918	0.5001	-273.3718	1790.9555	False
D	Y-Z	83.241	1.0	-1604.9068	1771.3888	False
E	F	130.2724	0.9806	-142.9612	403.506	False
	FANCY	809.6177	0.0	389.3719	1229.8635	True
E	G	310.8341	0.0245	17.2638	604.4044	True
E	H	366.8396	0.0056	52.6131	681.066	True
E	I	218.9501	0.509	-80.191	518.0912	False
E	J	110.6083	0.9997	-211.7693	432.9858	False
E	K	-149.8858	0.9983	-534.9151	235.1434	False
E	L	-214.8202	0.9923	-703.752	274.1116	False
E	M	-411.4685	0.7833	-1066.3254	243.3884	False
E	N	-124.8076	1.0	-1011.0009	761.3857	False
E	O	8.0421	1.0	-5796.0746	5812.1589	False
E	O-P	-99.9203	1.0	-1785.6666	1585.826	False
E	Q-R	553.6038	1.0	-1822.5117	2929.7194	False
E	S-T	170.0129	1.0	-1340.2532	1680.279	False
E	U-V	510.3298	0.9658	-503.2126	1523.8723	False
E	W	1620.6121	1.0	-4183.5046	7424.7288	False
E	W-X	848.8722	0.2734	-179.359	1877.1035	False
E	Y-Z	173.3214	1.0	-1512.4249	1859.0677	False
F	FANCY	679.3453	0.0	259.6418	1099.0488	True
F	G	180.5616	0.809	-112.2319	473.3552	False
F	H	236.5671	0.4468	-76.9337	550.068	False
F	I	88.6777	1.0	-209.7011	387.0565	False
F	J	-19.6641	1.0	-341.3345	302.0062	False
F	K	-280.1582	0.5179	-664.5956	104.2791	False
F	L	-345.0926	0.5801	-833.5584	143.3732	False
F	M	-541.7409	0.2689	-1196.25	112.7682	False
F	N	-255.08	1.0	-1141.0163	630.8563	False
F	O	-122.2303	1.0	-5926.3078	5681.8472	False
F	O-P	-230.1927	1.0	-1915.8039	1455.4185	False
F	Q-R	423.3314	1.0	-1952.6883	2799.3511	False
F	S-T	39.7405	1.0	-1470.3748	1549.8558	False
F	U-V	380.0574	0.999	-633.2603	1393.3751	False
F	W	1490.3397	1.0	-4313.7378	7294.4172	False
F	W-X	718.5998	0.6009	-309.4099	1746.6096	False
F	Y-Z	43.0489	1.0	-1642.5623	1728.6602	False
FANCY	G	-498.7837	0.0071	-932.0018	-65.5655	True
FANCY	H	-442.7782	0.0564	-890.2518	4.6955	False
FANCY	I	-590.6676	0.0003	-1027.68	-153.6553	True
FANCY	J	-699.0094	0.0	-1152.2442	-245.7747	True
FANCY	K	-959.5035	0.0	-1459.2455	-459.7616	True
FANCY	L	-1024.4379	0.0	-1608.0035	-440.8724	True
FANCY	M	-1221.0862	0.0	-1949.3252	-492.8472	True
FANCY	N	-934.4253	0.0546	-1876.1434	7.2928	False
FANCY	O	-801.5756	1.0	-6614.429	5011.2779	False
FANCY	O-P	-909.538	0.9438	-2625.124	806.0479	False
FANCY	Q-R	-256.0139	1.0	-2653.3915	2141.3638	False
FANCY	S-T	-639.6048	0.9962	-2183.1068	903.8971	False
FANCY	U-V	-299.2879	1.0	-1361.7203	763.1446	False
FANCY	W	810.9944	1.0	-5001.8591	6623.8478	False
FANCY	W-X	39.2545	1.0	-1037.1998	1115.7088	False
FANCY	Y-Z	-636.2964	0.9991	-2351.8823	1079.2896	False
G	H	56.0055	1.0	-275.3699	387.3809	False
G	I	-91.884	1.0	-408.9909	225.223	False
G	J	-200.2258	0.8607	-539.3404	138.8888	False

G	K	-460.7199	0.0068	-859.8677	-61.5721	True
	L	-525.6543	0.0271	-1025.78	-25.5285	True
G	M	-722.3026	0.0166	-1385.559	-59.0462	True
G	N	-435.6416	0.9751	-1328.0597	456.7764	False
G	O	-302.7919	1.0	-6107.8623	5502.2785	False
G	O-P	-410.7544	1.0	-2099.7813	1278.2726	False
G	Q-R	242.7698	1.0	-2135.6744	2621.2139	False
G	S-T	-140.8212	1.0	-1654.7482	1373.1059	False
G	U-V	199.4958	1.0	-819.4938	1218.4853	False
G	W	1309.778	1.0	-4495.2924	7114.8484	False
G	W-X	538.0382	0.9527	-495.5628	1571.6391	False
G	Y-Z	-137.5127	1.0	-1826.5397	1551.5143	False
H	I	-147.8895	0.9922	-484.2101	188.4311	False
H	J	-256.2313	0.5495	-613.3775	100.915	False
H	K	-516.7254	0.0017	-931.3019	-102.1489	True
H	L	-581.6598	0.0089	-1094.1834	-69.1361	True
H	M	-778.3081	0.0065	-1450.9623	-105.6538	True
H	N	-491.6471	0.9259	-1391.0718	407.7775	False
H	O	-358.7974	1.0	-6164.9491	5447.3542	False
H	O-P	-466.7599	1.0	-2159.4993	1225.9796	False
H	Q-R	186.7643	1.0	-2194.3177	2567.8463	False
H	S-T	-196.8267	1.0	-1714.8945	1321.2411	False
H	U-V	143.4903	1.0	-881.6411	1168.6217	False
H	W	1253.7725	1.0	-4552.3791	7059.9242	False
H	W-X	482.0327	0.9857	-557.6239	1521.6892	False
H	Y-Z	-193.5182	1.0	-1886.2576	1499.2212	False
I	J	-108.3418	0.9999	-452.2903	235.6067	False
I	K	-368.8359	0.1246	-772.0986	34.4268	False
I	L	-433.7703	0.2032	-937.1862	69.6457	False
I	M	-630.4186	0.0898	-1296.1595	35.3223	False
I	N	-343.7577	0.9986	-1238.0238	550.5085	False
I	O	-210.9079	1.0	-6016.2627	5594.4468	False
I	O-P	-318.8704	1.0	-2008.8745	1371.1337	False
I	Q-R	334.6538	1.0	-2044.4844	2713.7919	False
I	S-T	-48.9372	1.0	-1563.9544	1466.08	False
I	U-V	291.3798	1.0	-729.2287	1311.9882	False
I	W	1401.662	1.0	-4403.6928	7207.0168	False
I	W-X	629.9221	0.8261	-405.2749	1665.1191	False
I	Y-Z	-45.6287	1.0	-1735.6328	1644.3754	False
J	K	-260.4941	0.8039	-681.2823	160.2941	False
J	L	-325.4285	0.7823	-842.9896	192.1327	False
J	M	-522.0768	0.4014	-1198.5772	154.4236	False
J	N	-235.4159	1.0	-1137.7206	666.8889	False
J	O	-102.5661	1.0	-5909.1646	5704.0324	False
J	O-P	-210.5286	1.0	-1904.8001	1483.7429	False
J	Q-R	442.9956	1.0	-1939.1758	2825.1669	False
J	S-T	59.4046	1.0	-1460.3713	1579.1805	False
J	U-V	399.7216	0.9983	-627.9376	1427.3808	False
J	W	1510.0038	1.0	-4296.5947	7316.6023	False
J	W-X	738.264	0.5747	-303.8851	1780.413	False
J	Y-Z	62.7131	1.0	-1631.5584	1756.9846	False
K	L	-64.9344	1.0	-623.6738	493.8051	False
K	M	-261.5827	0.9992	-970.0833	446.9179	False
K	N	25.0782	1.0	-901.4605	951.617	False
K	O	157.928	1.0	-5652.4856	5968.3416	False
K	O-P	49.9655	1.0	-1657.3353	1757.2663	False
K	Q-R	703.4897	1.0	-1687.9661	3094.9454	False
K	S-T	319.8987	1.0	-1214.3891	1854.1865	False
K	U-V	660.2157	0.781	-388.7856	1709.217	False
K	W	1770.4979	0.9999	-4039.9157	7580.9115	False
K	W-X	998.7581	0.0971	-64.4423	2061.9584	False
K	Y-Z	323.2072	1.0	-1384.0936	2030.508	False
L	M	-196.6483	1.0	-966.5703	573.2737	False
L	N	90.0126	1.0	-884.2978	1064.3231	False
L	O	222.8623	1.0	-5595.3602	6041.0848	False
L	O-P	114.8999	1.0	-1618.7906	1848.5904	False
L	Q-R	768.424	0.9999	-1641.9426	3178.7907	False
L	S-T	384.8331	1.0	-1178.7672	1948.4334	False
L	U-V	725.15	0.6947	-366.2759	1816.5759	False
L	W	1835.4323	0.9999	-3982.7902	7653.6548	False
L	W-X	1063.6924	0.076	-41.3876	2168.7724	False
L	Y-Z	388.1416	1.0	-1345.549	2121.8321	False
M	N	286.6609	1.0	-780.6085	1353.9304	False
M	O	419.5106	1.0	-5414.9985	6254.0197	False
M	O-P	311.5482	1.0	-1476.0385	2099.1349	False

M	Q-R	965.0724	0.998	-1484.3461	3414.4908	False
M	S-T	581.4814	0.9994	-1041.6727	2204.6355	False
M	U-V	921.7983	0.3691	-253.3586	2096.9553	False
M	W	2032.0806	0.9996	-3802.4285	7866.5897	False
M	W-X	1260.3407	0.0239	72.4918	2448.1896	True
M	Y-Z	584.7899	0.9998	-1202.7968	2372.3765	False
N	O	132.8497	1.0	-5732.1309	5997.8303	False
N	O-P	24.8873	1.0	-1859.7791	1909.5536	False
N	Q-R	678.4114	1.0	-1842.7295	3199.5524	False
N	S-T	294.8205	1.0	-1434.668	2024.3089	False
N	U-V	635.1374	0.9782	-683.0032	1953.2781	False
N	W	1745.4197	1.0	-4119.5609	7610.4003	False
N	W-X	973.6798	0.5078	-355.7885	2303.1481	False
N	Y-Z	298.1289	1.0	-1586.5374	2182.7953	False
O	O-P	-107.9625	1.0	-6145.7103	5929.7854	False
O	Q-R	545.5617	1.0	-5720.1054	6811.2288	False
O	S-T	161.9707	1.0	-5829.1529	6153.0944	False
O	U-V	502.2877	1.0	-5383.2817	6387.8571	False
O	W	1612.5699	1.0	-6591.114	9816.2539	False
O	W-X	840.8301	1.0	-5047.2866	6728.9468	False
O	Y-Z	165.2792	1.0	-5872.4687	6203.0271	False
O-P	Q-R	653.5242	1.0	-2246.9161	3553.9644	False
O-P	S-T	269.9332	1.0	-1976.7382	2516.6046	False
O-P	U-V	610.2502	0.9999	-1337.5426	2558.043	False
O-P	W	1720.5324	1.0	-4317.2155	7758.2803	False
O-P	W-X	948.7925	0.9766	-1006.6839	2904.269	False
O-P	Y-Z	273.2417	1.0	-2094.9579	2641.4412	False
Q-R	S-T	-383.591	1.0	-3185.6825	2418.5005	False
Q-R	U-V	-43.274	1.0	-2611.9471	2525.3991	False
Q-R	W	1067.0082	1.0	-5198.6588	7332.6753	False
Q-R	W-X	295.2684	1.0	-2279.236	2869.7727	False
Q-R	Y-Z	-380.2825	1.0	-3280.7227	2520.1578	False
S-T	U-V	340.317	1.0	-1457.7542	2138.3881	False
S-T	W	1450.5992	1.0	-4540.5244	7441.7229	False
S-T	W-X	678.8594	0.9989	-1127.5324	2485.2511	False
S-T	Y-Z	3.3085	1.0	-2243.3629	2249.9799	False
U-V	W	1110.2822	1.0	-4775.2871	6995.8516	False
U-V	W-X	338.5424	1.0	-1078.9962	1756.081	False
U-V	Y-Z	-337.0085	1.0	-2284.8013	1610.7843	False
W	W-X	-771.7399	1.0	-6659.8566	5116.3768	False
W	Y-Z	-1447.2907	1.0	-7485.0386	4590.4571	False
W-X	Y-Z	-675.5509	0.9997	-2631.0273	1279.9256	False

Insight 🧠

The output shows the results of a Tukey HSD (honestly significant difference) multiple comparison test of means for groups labeled D to Z. The purpose of the test is to identify pairs of groups that have significantly different means. The test controls for the family-wise error rate (FWER) at 0.05, which means that the probability of making at least one type I error (rejecting a true null hypothesis) among all the comparisons is no more than 0.05.

The output shows the mean differences between pairs of groups along with p-values and confidence intervals. If the p-value is less than the significance level of 0.05, then we reject the null hypothesis that the means of the two groups are equal, and we conclude that the difference in means is statistically significant. If the p-value is greater than 0.05, then we fail to reject the null hypothesis, and we conclude that there is no evidence of a significant difference in means between the two groups.

Looking at the output, we see that only the comparison between group D and group FANCY has a p-value less than 0.05, indicating a significant difference in means. Therefore, we reject the null hypothesis that the means of these two groups are equal, and we conclude that there is a significant difference in means between group D and group FANCY. All other comparisons fail to reach significance, and we conclude that there is no evidence of a significant difference in means between those groups.

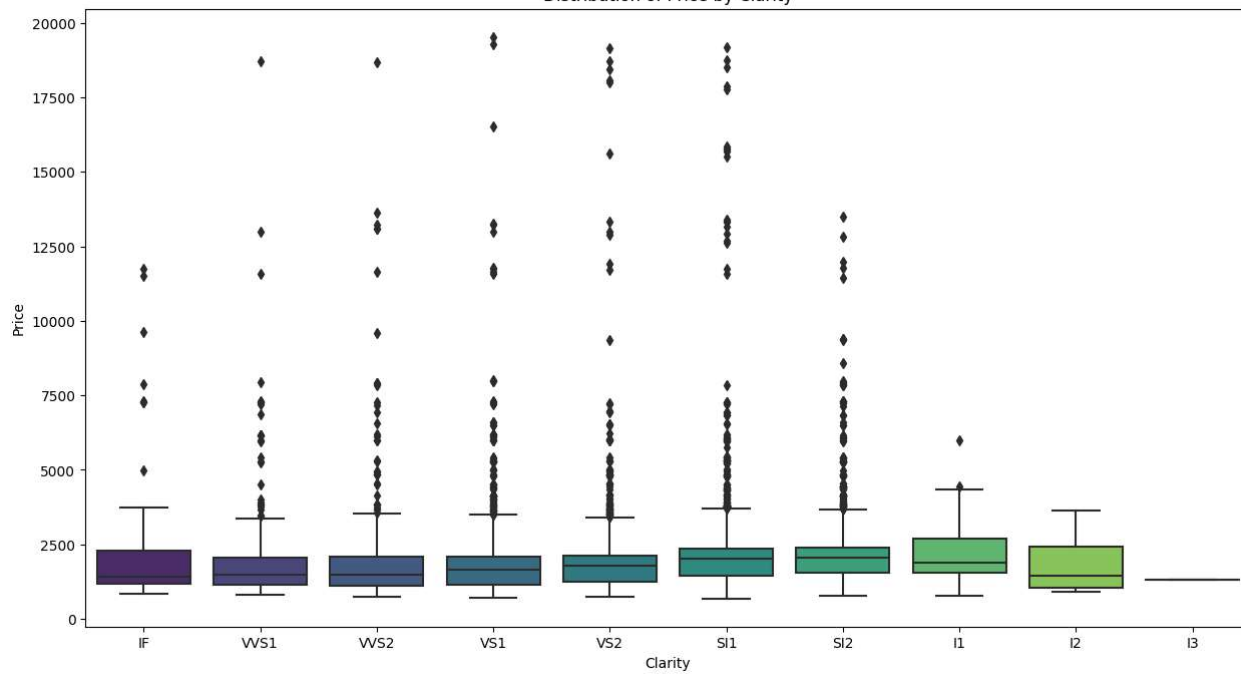
Test on Clarity feature

```
In [ ]: df["Clarity"].unique()

Out[ ]: ['SI2', 'VS1', 'VS2', 'SI1', 'VS2', 'VS1', 'I3', 'I1', 'IF', 'I2']
Categories (11, object): ['FL', 'I1', 'I2', 'I3', ..., 'VS1', 'VS2', 'VS1', 'VS2']

In [ ]: groupby_boxplot(
df,
"Clarity",
"Price",
order=["IF", "VS1", "VS2", "VS1", "VS2", "SI1", "SI2", "I1", "I2", "I3"],
)
```

Distribution of Price by Clarity



```
In [ ]: # one-way ANOVA test
model = sm.formula.ols("Price ~ Clarity", data=df).fit()
table = sm.stats.anova_lm(model, typ=2)
table
```

```
/mnt/d/Code/Python/AI/Diamond/env/lib/python3.10/site-packages/statsmodels/base/model.py:1871: ValueWarning: covariance of constraints does not have full rank. The number of constraints is 10, but rank is 1
warnings.warn('covariance of constraints does not have full ')
```

```
Out[ ]:
```

	sum_sq	df	F	PR(>F)
Clarity	2.354029e+06	10.0	0.087411	0.767505
Residual	1.584590e+10	5884.0	NaN	NaN

Insight 💡

Based on the ANOVA results provided, we can conclude that there is no significant difference in the mean prices of diamonds among the different clarity groups ($F=0.087411$, $p\text{-value}=0.767505$). Therefore, the clarity of the diamond does not have a statistically significant effect on the price of diamonds. However, it is always recommended to perform post-hoc tests to further investigate the relationships between variables.

```
In [ ]: # Tukey's HSD test
tukey = pairwise_tukeyhsd(endog=df["Price"], groups=df["Clarity"], alpha=0.05)
tukey.summary()
```

Out []:

group1	group2	meandiff	p-adj	lower	upper	reject
I1	I2	-352.9644	1.0	-2638.566	1932.6371	False
I1	I3	-883.3495	1.0	-6146.3538	4379.6547	False
I1	IF	106.7843	1.0	-927.7797	1141.3484	False
I1	SI1	53.5251	1.0	-811.4078	918.458	False
I1	SI2	136.3785	1.0	-733.6073	1006.3642	False
I1	VS1	-215.1844	0.9988	-1083.3	652.9312	False
I1	VS2	-180.047	0.9997	-1045.9952	685.9012	False
I1	VVS1	-335.5564	0.9735	-1225.3026	554.1897	False
I1	VVS2	-299.7924	0.9868	-1179.8842	580.2995	False
I2	I3	-530.3851	1.0	-6139.7821	5079.012	False
I2	IF	459.7488	0.9997	-1739.4429	2658.9404	False
I2	SI1	406.4896	0.9999	-1718.1822	2531.1613	False
I2	SI2	489.3429	0.9993	-1637.3908	2616.0766	False
I2	VS1	137.78	1.0	-1988.1894	2263.7494	False
I2	VS2	172.9175	1.0	-1952.1678	2298.0028	False
I2	VVS1	17.408	1.0	-2117.4853	2152.3013	False
I2	VVS2	53.1721	1.0	-2077.7157	2184.0599	False
I3	IF	990.1338	0.9999	-4235.9242	6216.1919	False
I3	SI1	936.8746	0.9999	-4258.2644	6132.0136	False
I3	SI2	1019.728	0.9998	-4176.2547	6215.7106	False
I3	VS1	668.1651	1.0	-4527.5047	5863.8349	False
I3	VS2	703.3025	1.0	-4492.0056	5898.6107	False
I3	VVS1	547.7931	1.0	-4651.5346	5747.1208	False
I3	VVS2	583.5571	1.0	-4614.1272	5781.2414	False
IF	SI1	-53.2592	1.0	-653.741	547.2225	False
IF	SI2	29.5941	1.0	-578.1431	637.3313	False
IF	VS1	-321.9687	0.8047	-927.0259	283.0884	False
IF	VS2	-286.8313	0.8894	-888.7746	315.1119	False
IF	VVS1	-442.3408	0.4552	-1078.0432	193.3617	False
IF	VVS2	-406.5767	0.5501	-1028.6949	215.5415	False
SI1	SI2	82.8534	0.9713	-134.2429	299.9496	False
SI1	VS1	-268.7095	0.002	-478.1859	-59.2332	True
SI1	VS2	-233.5721	0.0086	-433.8767	-33.2675	True
SI1	VVS1	-389.0815	0.0007	-675.2701	-102.893	True
SI1	VVS2	-353.3175	0.0005	-607.9099	-98.7251	True
SI2	VS1	-351.5629	0.0001	-581.0112	-122.1145	True
SI2	VS2	-316.4255	0.0003	-537.5321	-95.3188	True
SI2	VVS1	-471.9349	0.0	-773.0496	-170.8202	True
SI2	VVS2	-436.1708	0.0	-707.4336	-164.9081	True
VS1	VS2	35.1374	1.0	-178.4925	248.7673	False
VS1	VVS1	-120.372	0.9564	-416.0402	175.2961	False
VS1	VVS2	-84.608	0.9918	-349.8118	180.5959	False
VS2	VVS1	-155.5094	0.7949	-444.752	133.7331	False
VS2	VVS2	-119.7454	0.9045	-377.7661	138.2753	False
VVS1	VVS2	35.7641	1.0	-293.4116	364.9397	False

Insight 🔍

The table presents the results of a multiple comparison of means test using Tukey's HSD (honestly significant difference) method with a family-wise error rate (FWER) of 0.05. The test was performed on a dataset consisting of observations from several groups labeled as I1, I2, I3, IF, SI1, SI2, VS1, VS2, VVS1, and VVS2.

The table shows the mean differences between pairs of groups along with their corresponding p-values, confidence intervals, and a column indicating whether the null hypothesis of no difference between the two groups was rejected or not.

Based on the results, we can conclude that there are no significant differences between the mean values of any pair of groups at a significance level of 0.05. All the p-values are above 0.05, and none of the null hypotheses were rejected. Therefore, we fail to reject the null hypothesis that there is no difference in means between any pair of groups.

Summary 📄

All categorical data are ordinal except Shape This are the list of ordinal data (best to worst)

- Polish: ['EX', 'VG', 'GD', 'FR']
- Symmetry: ['EX', 'VG', 'GD', 'FR']
- Fluorescence: ['VSL', 'VS', 'ST', 'SL', 'F', 'M', 'N']
- Cut: ['EX', 'VG', 'GD', 'FR']
- Colour: ['D', 'E', 'F', 'FANCY', ..., 'U-V', 'W', 'W-X', 'Y-Z']
- Clarity: ['IF', 'VVS1', 'VVS2', 'VS1', 'VS2', 'SI1', 'SI2', 'I1', 'I2', 'I3']

Chi-square test

In this test, we will determine weather 2 ordinal features are correlate to each other or not

```
In [ ]: def chi_square_test(feature_1, feature_2, alpha=0.05):
# Create a contingency table of the variables
contingency_table = pd.crosstab(feature_1, feature_2)

# Chi-square test to test for a relationship between variables
chi2_stat, p_value, dof, expected_freq = chi2_contingency(contingency_table)

# Print the results
print(
    f"Chi-square test results for the relationship between {feature_1.name} and {feature_2.name}:"
)
print(f"Chi-square statistic: {chi2_stat:.2f}")
print(f"p-value: {p_value:.5f}")
if p_value < alpha:
    print(
```

```
    )
    else:
        print(
            f"There is not enough evidence to suggest a significant relationship between {feature_1.name} and {feature_2.name}."
        )
```

This is a function called `chi_square_test` that performs a chi-square test of independence between two categorical variables and prints the results.

Test on all ordinal features

```
In [ ]: chi_square_test(df.Clarity, df.Colour)

Chi-square test results for the relationship between Clarity and Colour:
Chi-square statistic: 581.48
p-value: 0.00000
There is a significant relationship between Clarity and Colour.

In [ ]: chi_square_test(df.Clarity, df.Cut)

Chi-square test results for the relationship between Clarity and Cut:
Chi-square statistic: 169.48
p-value: 0.00000
There is a significant relationship between Clarity and Cut.

In [ ]: chi_square_test(df.Clarity, df.Polish)

Chi-square test results for the relationship between Clarity and Polish:
Chi-square statistic: 65.89
p-value: 0.00004
There is a significant relationship between Clarity and Polish.

In [ ]: chi_square_test(df.Clarity, df.Symmetry)

Chi-square test results for the relationship between Clarity and Symmetry:
Chi-square statistic: 184.42
p-value: 0.00000
There is a significant relationship between Clarity and Symmetry.

In [ ]: chi_square_test(df.Clarity, df.Fluorescence)

Chi-square test results for the relationship between Clarity and Fluorescence:
Chi-square statistic: 111.00
p-value: 0.00001
There is a significant relationship between Clarity and Fluorescence.

In [ ]: chi_square_test(df.Colour, df.Cut)

Chi-square test results for the relationship between Colour and Cut:
Chi-square statistic: 132.41
p-value: 0.00000
There is a significant relationship between Colour and Cut.

In [ ]: chi_square_test(df.Colour, df.Polish)

Chi-square test results for the relationship between Colour and Polish:
Chi-square statistic: 83.79
p-value: 0.01196
There is a significant relationship between Colour and Polish.

In [ ]: chi_square_test(df.Colour, df.Symmetry)

Chi-square test results for the relationship between Colour and Symmetry:
Chi-square statistic: 316.72
p-value: 0.00000
There is a significant relationship between Colour and Symmetry.

In [ ]: chi_square_test(df.Colour, df.Fluorescence)

Chi-square test results for the relationship between Colour and Fluorescence:
Chi-square statistic: 573.55
p-value: 0.00000
There is a significant relationship between Colour and Fluorescence.

In [ ]: chi_square_test(df.Cut, df.Polish)

Chi-square test results for the relationship between Cut and Polish:
Chi-square statistic: 144.65
p-value: 0.00000
There is a significant relationship between Cut and Polish.

In [ ]: chi_square_test(df.Cut, df.Symmetry)

Chi-square test results for the relationship between Cut and Symmetry:
Chi-square statistic: 644.91
p-value: 0.00000
There is a significant relationship between Cut and Symmetry.

In [ ]: chi_square_test(df.Cut, df.Fluorescence)

Chi-square test results for the relationship between Cut and Fluorescence:
Chi-square statistic: 22.02
p-value: 0.23102
There is not enough evidence to suggest a significant relationship between Cut and Fluorescence.

In [ ]: chi_square_test(df.Polish, df.Symmetry)

Chi-square test results for the relationship between Polish and Symmetry:
Chi-square statistic: 776.17
p-value: 0.00000
There is a significant relationship between Polish and Symmetry.

In [ ]: chi_square_test(df.Polish, df.Fluorescence)

Chi-square test results for the relationship between Polish and Fluorescence:
Chi-square statistic: 41.95
p-value: 0.00112
There is a significant relationship between Polish and Fluorescence.

In [ ]: chi_square_test(df.Symmetry, df.Fluorescence)

Chi-square test results for the relationship between Symmetry and Fluorescence:
Chi-square statistic: 50.97
p-value: 0.00005
There is a significant relationship between Symmetry and Fluorescence.
```

Insight 🧡

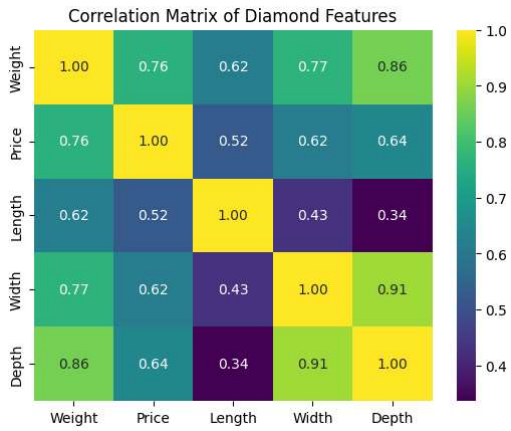
Using chi-square test we found out that all categorical data are correlated to each other except Cut and Fluorescence

Correlation Matrix of Diamond Features

In this test, we will determine correlation relation for all numeric features

```
In [ ]: sns.heatmap(df.corr(numeric_only=True), annot=True, fmt=".2f", cmap="viridis")
plt.title("Correlation Matrix of Diamond Features")
```

```
plt.show()
```



Insight 🔦

Using Pearson correlation coefficient we found out that all numeric data positive correlate to each other

Conclusion 🎉

In this project, we found that all categorical data are ordinal except Shape.

This are the list of ordinal data (best to worst)

- Polish : ['EX', 'VG', 'GD', 'FR']
- Symmetry : ['EX', 'VG', 'GD', 'FR']
- Fluorescence : ['VSL', 'VS', 'ST', 'SL', 'F', 'M', 'N']
- Cut : ['EX', 'VG', 'GD', 'FR']
- Colour : ['D', 'E', 'F', 'FANCY', ..., 'U-V', 'W', 'W-X', 'Y-Z']
- Clarity : ['IF', 'VVS1', 'VVS2', 'VS1', 'VS2', 'SI1', 'SI2', 'I1', 'I2', 'I3']

All categorical date are correlate to each other except Cut and Fluorescence.

And all numeric data positive correlate to each other